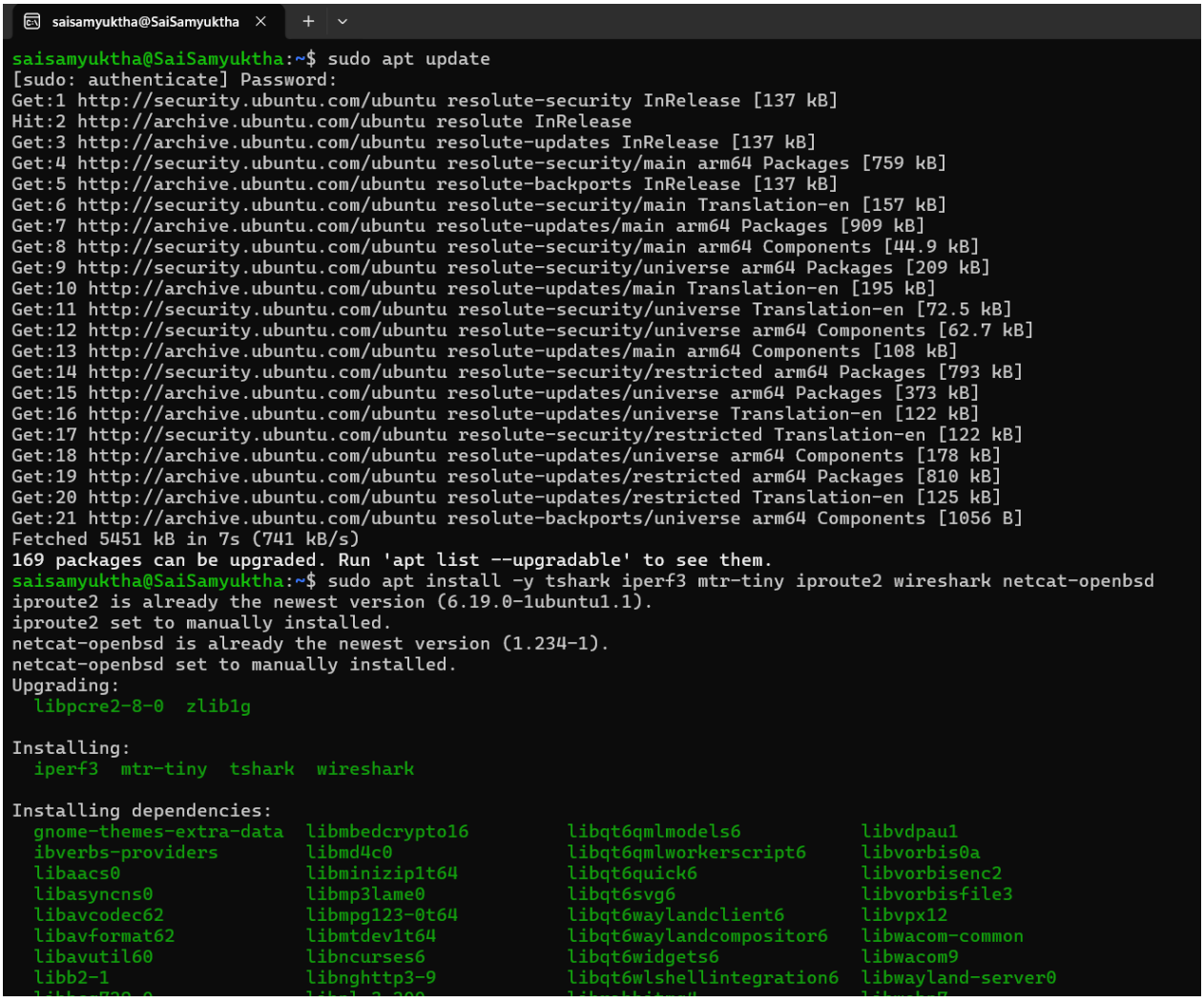
CNNP – Skill 2

THREE-POINT CAPTURE – Analyze packet captures from three points in a network path to understand packet flow and identify performance differences.

STAGE 1 — Install tools

Open Ubuntu and type the following commands in the Ubuntu terminal:

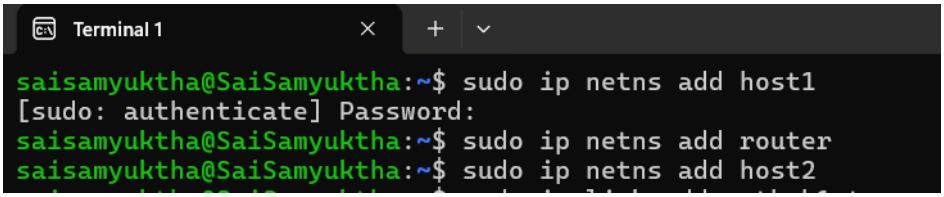
sudo apt update
sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd



STAGE 2 — Build the network

Open Ubuntu again, rename that tab as **Terminal 1** and run all of these one by one:

sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2



sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2

```
saisamyuktha@SaiSamyuktha:~$ sudo ip link add veth-h1 type veth peer name veth-r1
saisamyuktha@SaiSamyuktha:~$ sudo ip link add veth-r2 type veth peer name veth-h2
```

sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2

```
saisamyuktha@SaiSamyuktha:~$ sudo ip link set veth-h1 netns host1
saisamyuktha@SaiSamyuktha:~$ sudo ip link set veth-r1 netns router
saisamyuktha@SaiSamyuktha:~$ sudo ip link set veth-r2 netns router
saisamyuktha@SaiSamyuktha:~$ sudo ip link set veth-h2 netns host2
```

STAGE 3 — Configure IP addresses (Note: In Terminal 1 only)

host1:

sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up

```
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host1 ip link set veth-h1 up
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host1 ip link set lo up
```

router:

sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up

```
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
```

host2:

sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up

```
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
```

STAGE 4 — Enable routing

sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1

```
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
```

STAGE 5 — Test connectivity

sudo ip netns exec host1 ping -c 3 10.0.2.2

```
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.101 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.150 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.189 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2035ms
rtt min/avg/max/mdev = 0.101/0.146/0.189/0.036 ms
```

You must see: 3 packets transmitted, 3 received, 0% packet loss

STAGE 6 — Add delay and loss

Verify:

sudo ip netns exec router tc qdisc show dev veth-r2

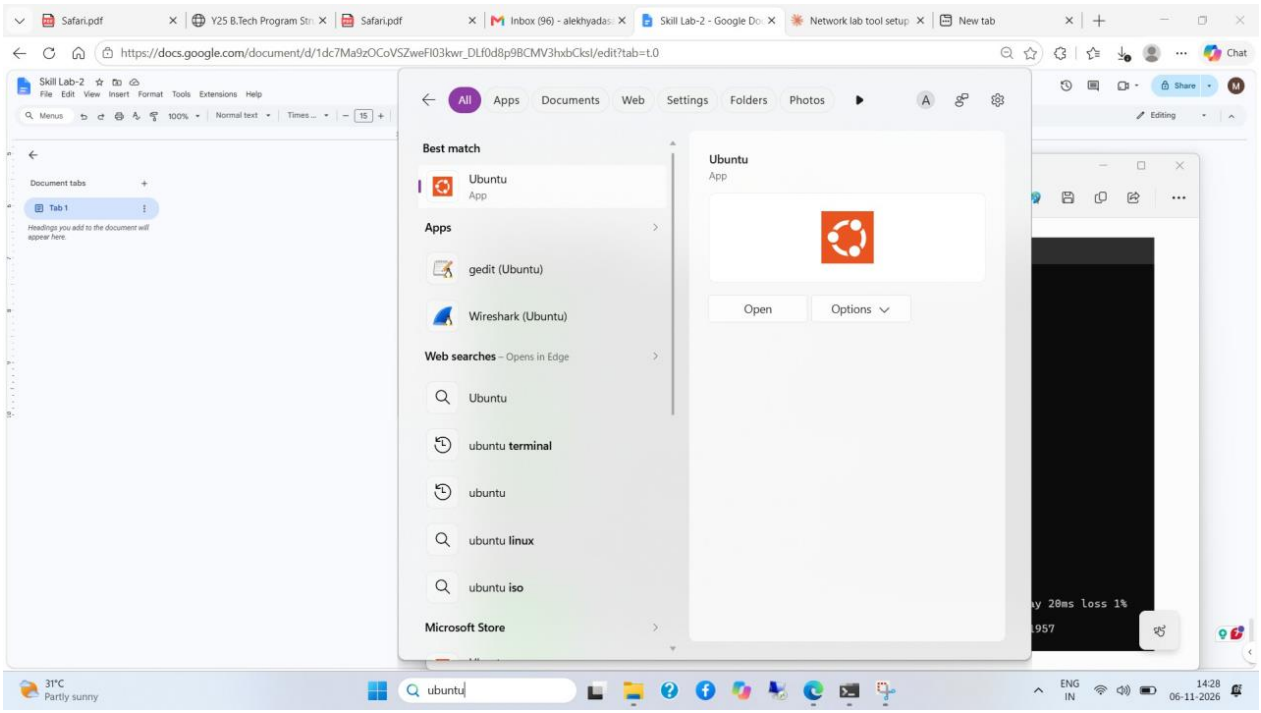
```
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 100ms
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 9 limit 1000 delay 100ms seed 14507913640732205493
```

You must see: netem delay 100ms

STAGE 7 — Open 4 terminals

You need 4 terminals open now.

Terminal 1 — already open (used above)
Terminal 2 — open new (by going to the Ubuntu option)
Terminal 3 — open new (by going to the Ubuntu option)
Terminal 4 — open new (by going to the Ubuntu option)



STAGE 8 — Start three captures simultaneously

Terminal 1 — capture at host1

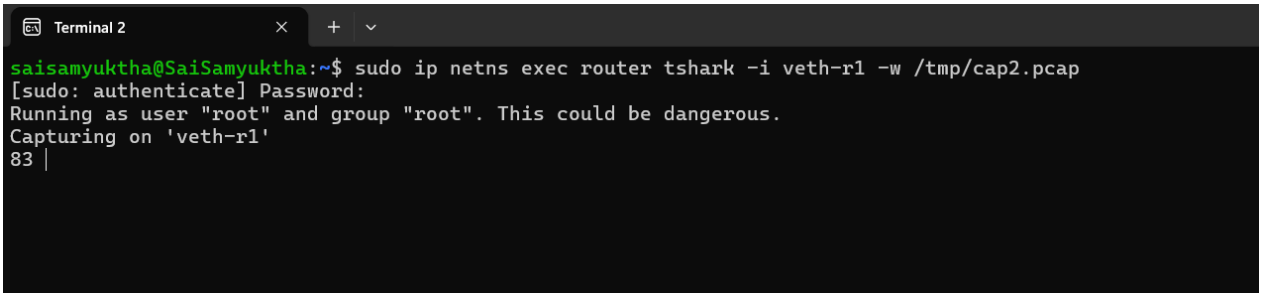
sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap

```
Terminal 1
saisamyuktha@SaiSamyuktha:~$ sudo ip netns add host1
[sudo: authenticate] Password:
saisamyuktha@SaiSamyuktha:~$ sudo ip netns add router
saisamyuktha@SaiSamyuktha:~$ sudo ip netns add host2
saisamyuktha@SaiSamyuktha:~$ sudo ip link add veth-h1 type veth peer name veth-r1
saisamyuktha@SaiSamyuktha:~$ sudo ip link add veth-r2 type veth peer name veth-h2
saisamyuktha@SaiSamyuktha:~$ sudo ip link set veth-h1 netns host1
saisamyuktha@SaiSamyuktha:~$ sudo ip link set veth-r1 netns router
saisamyuktha@SaiSamyuktha:~$ sudo ip link set veth-r2 netns router
saisamyuktha@SaiSamyuktha:~$ sudo ip link set veth-h2 netns host2
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host1 ip link set veth-h1 up
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host1 ip link set lo up
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.101 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.150 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.189 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2035ms
rtt min/avg/max/mdev = 0.101/0.146/0.189/0.036 ms
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 100ms
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 9 limit 1000 delay 100ms seed 14507913640732205493
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
83
```

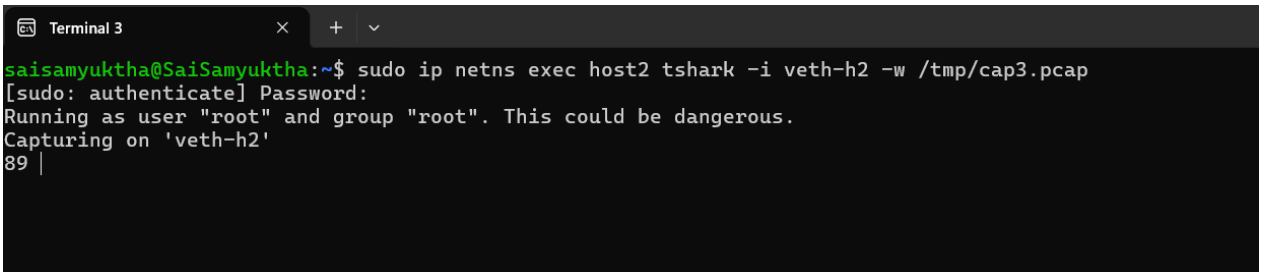
Terminal 2 — capture at router

sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap



Terminal 3 — capture at host2

sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap



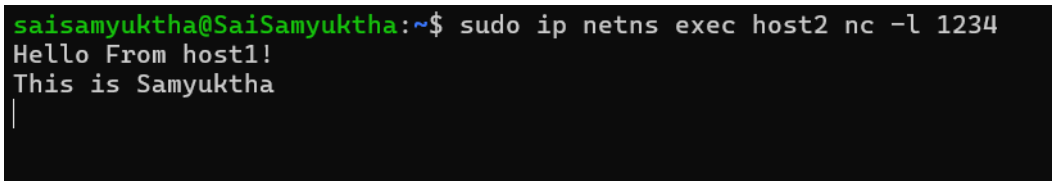
All three show: Capturing on 'veth-h1'

Leave all three running.

STAGE 9 — Send real message from host1 to host2

Go to Terminal 4. First start the receiver on host2:

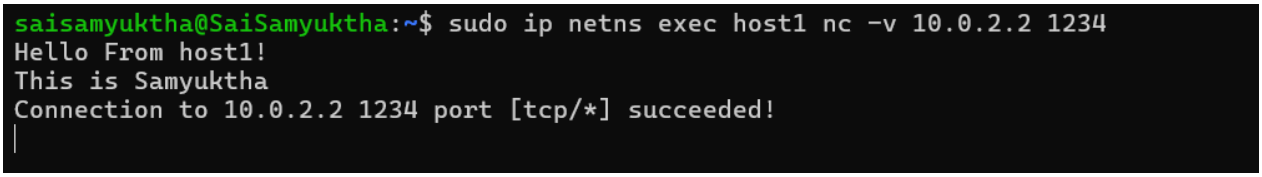
sudo ip netns exec host2 nc -l 1234



Open a **5th terminal**.

Connect from host1:

sudo ip netns exec host1 nc 10.0.2.2 1234



Now type this message and press Enter:

Hello from host1!

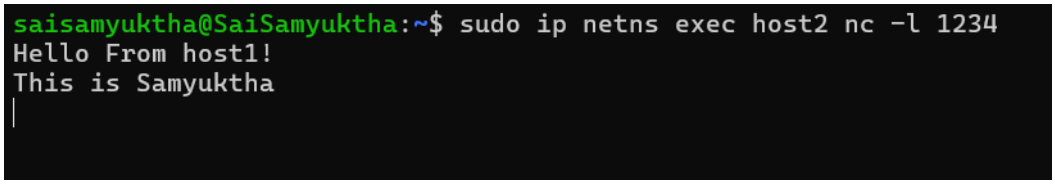
Type more messages:

This is Alekhya
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router

STAGE 10 — What you see in real time

Terminal 4 (host2) shows your message arriving:

Hello from host1!
This is Alekhya
Computer Networks Lab
Three Point Capture Experiment
Message travelling through router



Terminals 1, 2, 3 (tshark) show packet counts increasing:

Capturing on 'veth-h1'
10 packets
24 packets
31 packets

STAGE 12 — Fix permissions

sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap

STAGE 13 — Check TTL

tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

Expected:

cap1 → 64
cap2 → 64
cap3 → 63

```
saisamyuktha@SaiSamyuktha:~$ tshark -r /tmp/cap1.pcap -Y "ip.src == 10.0.1.1 && ip.dst == 10.0.2.2" -T fields -e ip.ttl
64
64
64
64
64
64
64
64
64
64
64
64
64
64
64
64
64
64
64
64
64
```

```
saisamyuktha@SaiSamyuktha:~$ tshark -r /tmp/cap2.pcap -Y "ip.src == 10.0.1.1 && ip.dst == 10.0.2.2" -T fields -e ip.ttl
64
64
64
64
64
64
64
64
64
64
64
```

```
saisamyuktha@SaiSamyuktha:~$ tshark -r /tmp/cap3.pcap -Y "ip.src == 10.0.1.1 && ip.dst == 10.0.2.2" -T fields -e ip.ttl
63
63
63
63
63
63
63
63
63
63
63
```

STAGE 14 — Read your actual message from the capture

tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e data.text 2>/dev/null

You will see your actual message text extracted from the packet:

Hello from host1!
This is Alekhya
Computer Networks Lab
Three Point Capture Experiment
Message travelling through router

```
saisamyuktha@SaiSamyuktha:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload | xxd -r -p
Hello from host1!
This is Alekhya
Computer Networks Lab
Three Point Capture Experiment



Hello from host1!
This is Samyuktha
Computer Networks Lab
Three Point Capture Experiment
Hello From host1!
This is Samyuktha
saisamyuktha@SaiSamyuktha:~$
```

Check delay:

tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null

tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null

tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null

Difference between cap1 and cap3 ≈ 20ms

Check retransmissions:

tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l

```
saisamyuktha@SaiSamyuktha:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
19
saisamyuktha@SaiSamyuktha:~$ tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
19
saisamyuktha@SaiSamyuktha:~$ tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
19
saisamyuktha@SaiSamyuktha:~$
```

cap3 shows highest count — loss happened between router and host2.

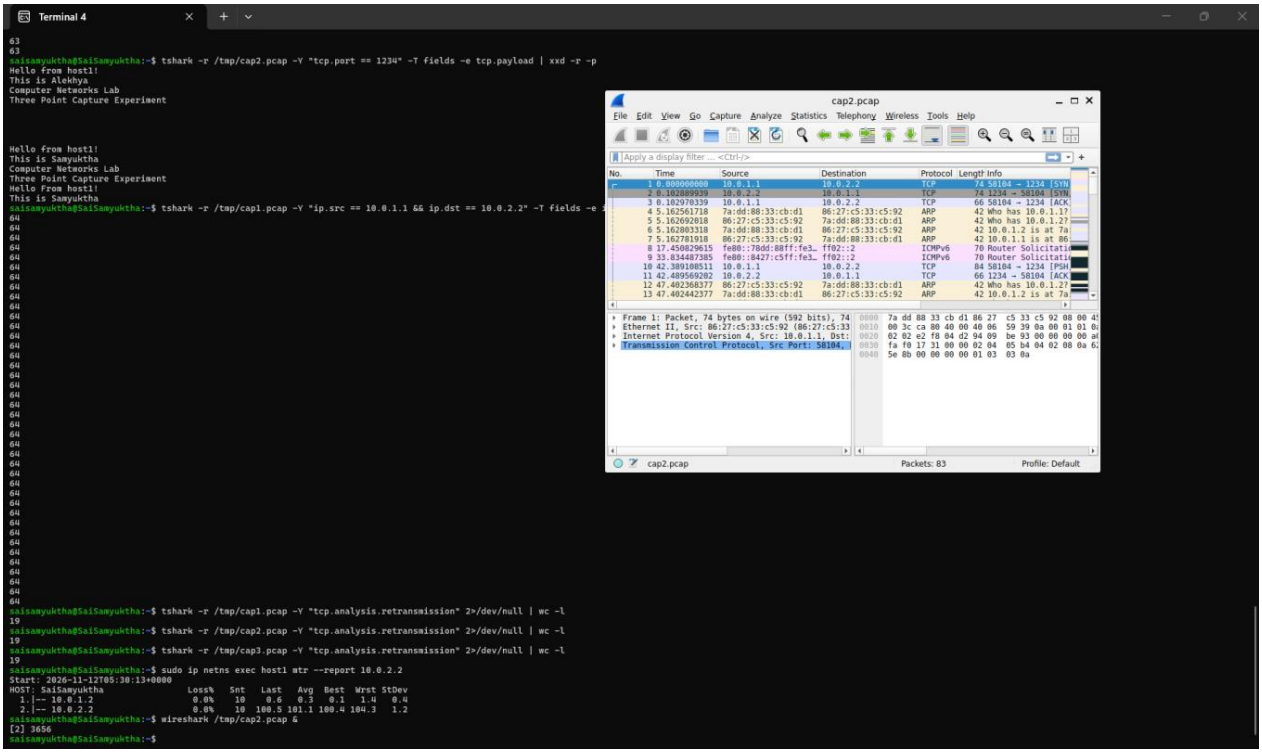
Check RTT:

sudo ip netns exec host1 mtr --report 10.0.2.2

```
saisamyuktha@SaiSamyuktha:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
19
saisamyuktha@SaiSamyuktha:~$ tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
19
saisamyuktha@SaiSamyuktha:~$ tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
19
saisamyuktha@SaiSamyuktha:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-12T05:30:13+0000
HOST: SaiSamyuktha                Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.2                   0.0%    10    0.6   0.3   0.1   1.4   0.4
  2.|-- 10.0.2.2                   0.0%    10  100.5 101.1 100.4 104.3   1.2
saisamyuktha@SaiSamyuktha:~$
```

TERMINAL 4 — Open Wireshark

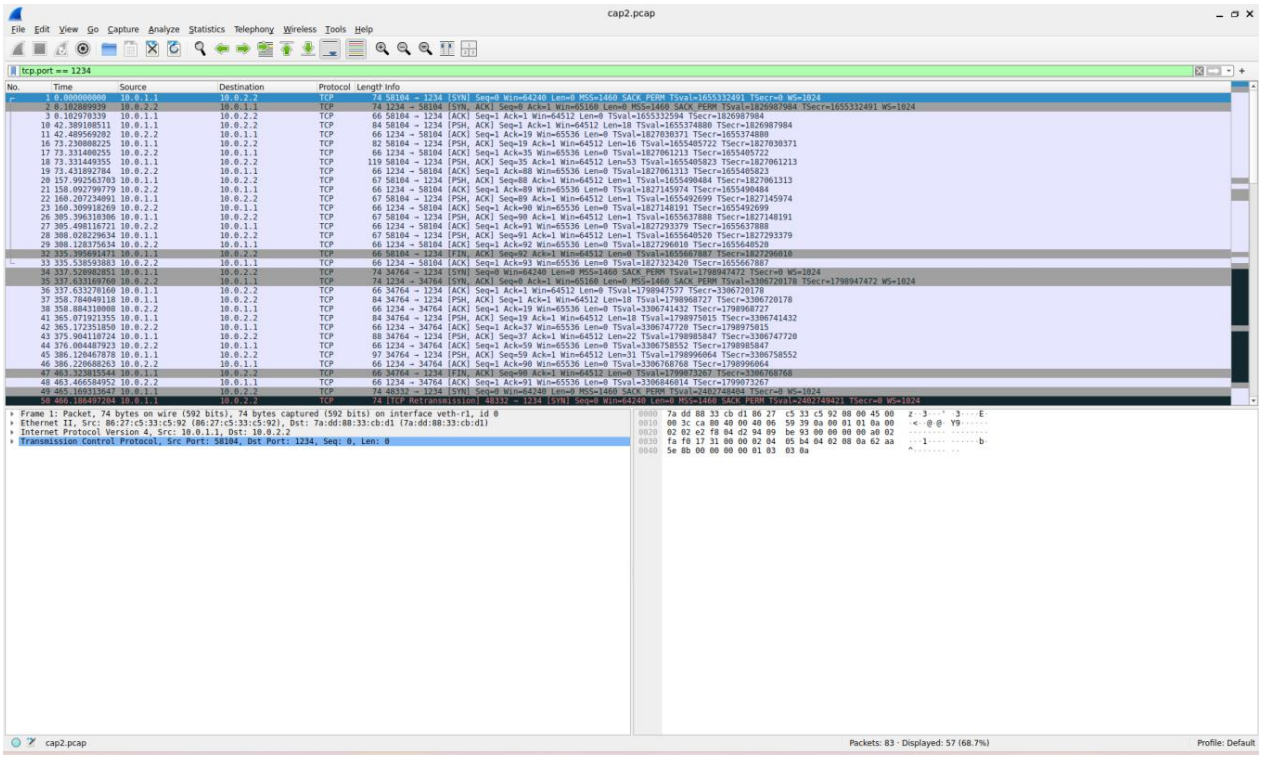
wireshark /tmp/cap2.pcap &



Inside Wireshark — filter your message only:

tcp.port == 1234

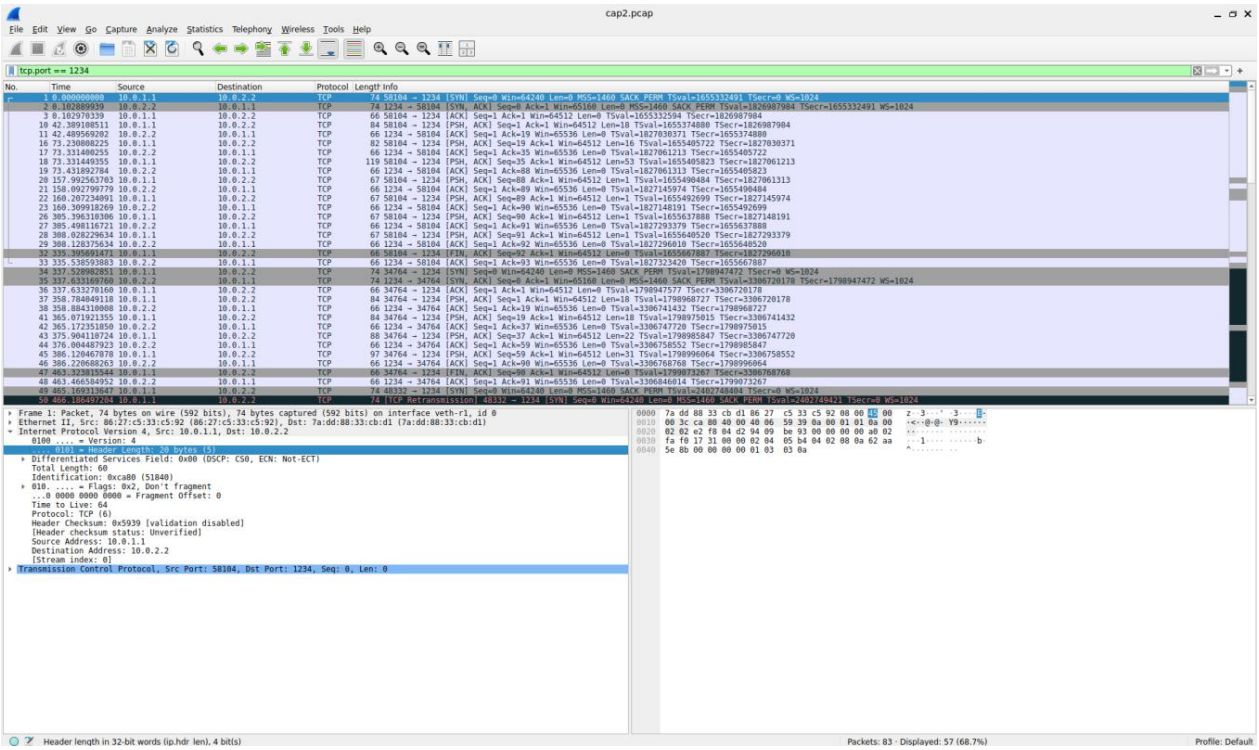
Click any packet → expand middle pane → you will see:



▾ Ethernet II

▾ Internet Protocol Version 4

Time to Live: 64 ← TTL visible here



▾ Transmission Control Protocol

▾ Data

Hello from host1! ← your actual message visible here



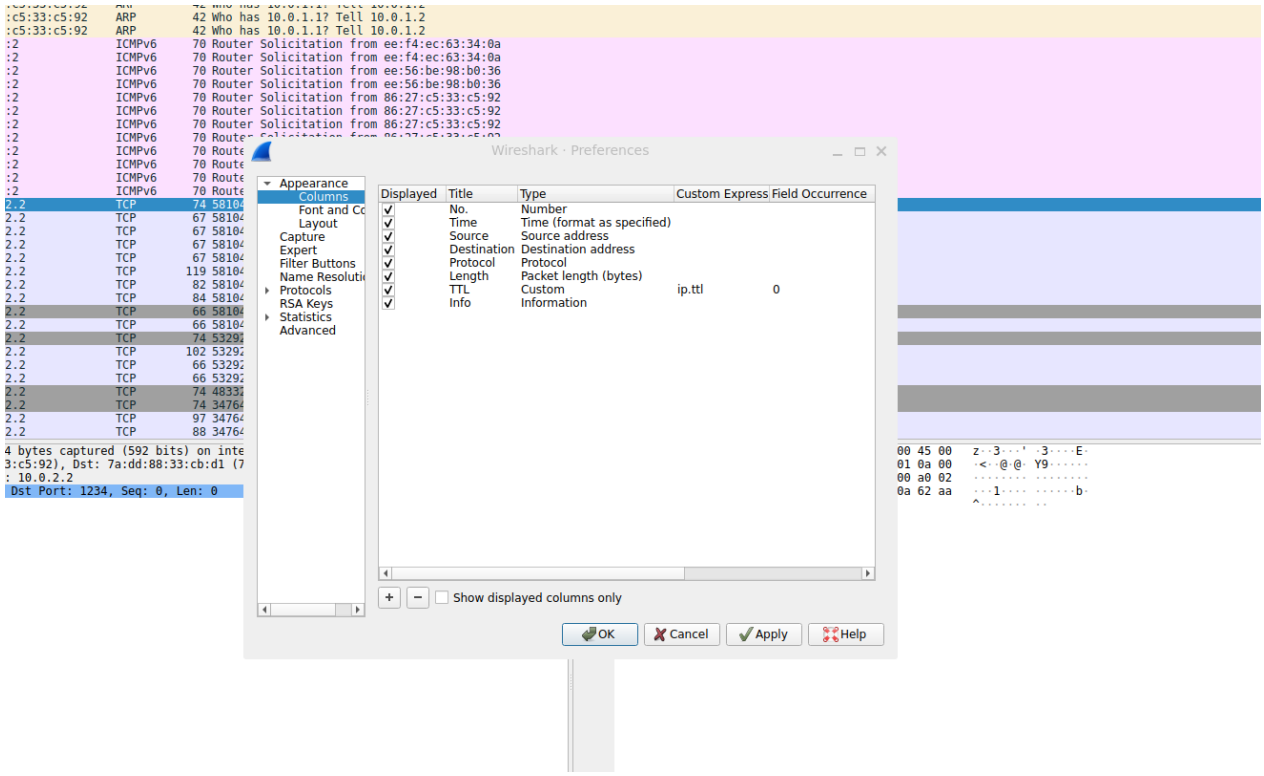
Merge all 3 captures:

File → Merge → select cap1.pcap

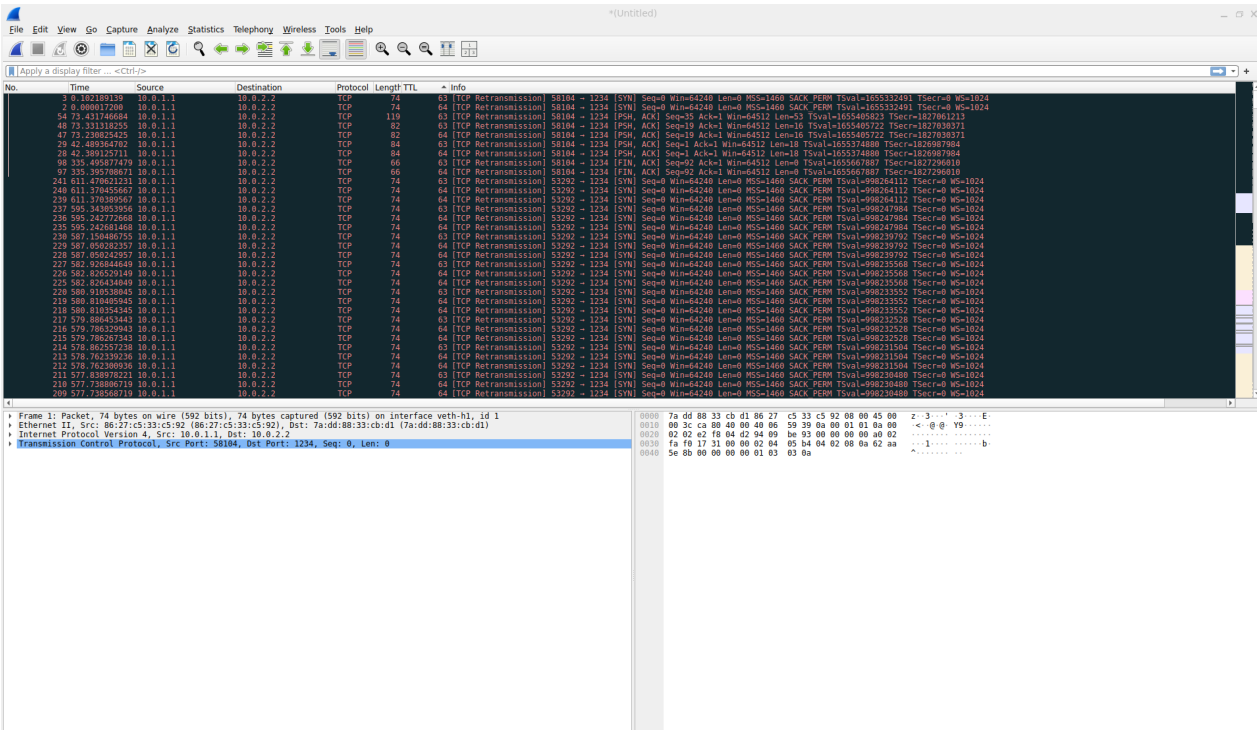
File → Merge → select cap3.pcap

Add TTL column:

Right-click column header
→ Column Preferences
→ click +
→ Title: TTL
→ Type: Custom
→ Fields: ip.ttl
→ OK

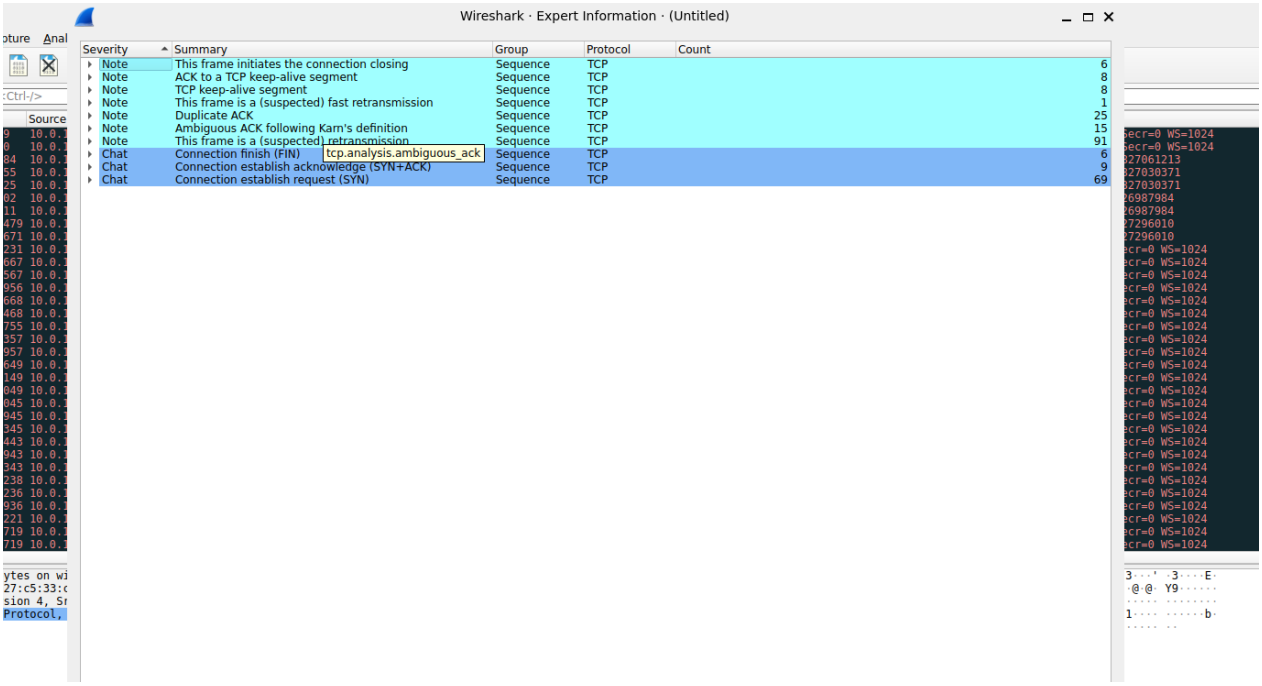


Students can now see TTL = 64 at points (1) and (2) and TTL = 63 at point (3).



Show packet loss:

Analyze → Expert Information



CLEAN UP — Terminal 4

sudo ip netns del host1
sudo ip netns del router
sudo ip netns del host2

Verify:

sudo ip netns list

Must return blank.

